Universidad Internacional del Ecuador (UIDE)
Computer Science School
Network Communication Technology
Title: *Phase 2 Packet Tracer Network Prototype*
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Link to the GitHub project
https://github.com/TownECreator/correos-integrador
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1. **Updated Architecture**

Diagram Updates and Justification of Changes The initial architecture proposed in Phase 1 has been updated to reflect the strict addressing constraints and the physical simulation environment modeled in Cisco Packet Tracer. The primary changes include:

- **Addressing Scheme:** The conceptual WAN link from Phase 1 (which initially used 192.168.100.0/24) has been redesigned to strictly adhere to the project's 10.10.0.0/16 boundary constraint. A /30 subnet (10.10.254.0/30) is now utilized for the inter-site WAN connection, ensuring zero IP waste and maintaining full compliance with the new addressing requirements.
- **Hardware Representation:** To simulate the target OpenWrt deployment accurately before the Week 10 hardware distribution, Cisco ISR 4331 routers have been selected as the edge devices. Consequently, interface labels in the topology have been updated to match the physical simulation (Gig0/0/0 for the LAN and Gig0/0/1 for the WAN).

Implementation Status Matrix To track the project's progress towards the final physical integration, the architectural elements have been divided into currently active simulated components and planned features:

Implemented Elements (Active in Packet Tracer Prototype):

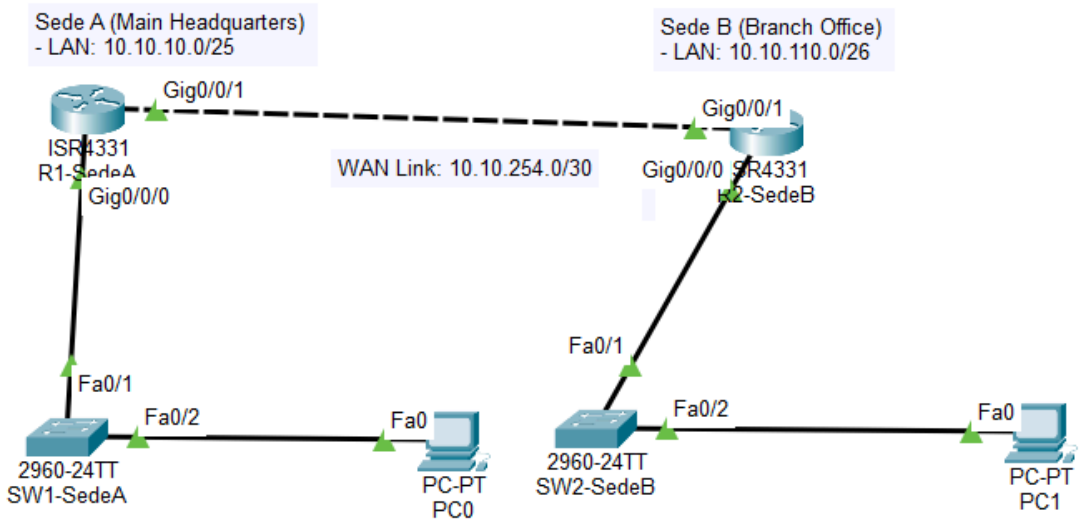
- **Edge Routers:** Cisco ISR 4331 devices representing the future OpenWrt/Raspberry Pi deployment at both sites.
- **Distribution Switches:** Cisco Catalyst 2960 switches providing initial Layer 2 connectivity.
- **End-User Simulation:** Client PCs representing the active User networks at Sede A and Sede B.
- **Layer 3 Addressing:** Static IP assignments on router interfaces utilizing the planned VLSM model.
- **Physical Connectivity:** Active WAN link (cross-over cable) between sites and active LAN links (straight-through cables) to switches and hosts.
- **DHCP:** pools on both routers for the active user networks.
- **Static Intersite Routing:** static routes over the 10.10.254.0/30 link.
- **Secure Administration:** SSH v2 with local authentication on routes and switches.
- **Switch Management:** management IP on VLAN1 on both switches.

Planned Elements (Pending for Week 10+ and Hardware Deployment):

- **VLAN Segmentation**: Implementation of isolated broadcast domains (VLAN 10 for Users, VLAN 20 for Servers, VLAN 30 for Guests, VLAN 99 for Management).

- **Router-on-a-Stick (802.1Q)**: Trunking configuration on switches and subinterfaces on routers to allow inter-VLAN routing.
- **Redundancy & Loop Prevention:** EtherChannel (LACP) and RSTP configuration between multiple Catalyst switches.
- **Dynamic Routing:** Full OSPFv2 deployment across the WAN link to dynamically share the internal subnets.
- **Application Servers:** Deployment of the proprietary Go (Email) and FastAPI (Auth) services within the secured Server segment (VLAN 20).
- **Hardware Migration:** Translating the Cisco CLI configuration logic into UCI (Unified Configuration Interface) for the physical OpenWrt devices.
- **Network Address Translation (NAT):** Configuration of PAT (overload) on the edge routers so that internal 10.10.0.0/16 hosts can reach external networks through a single public-facing address. On OpenWrt this will be implemented with masquerading (masq) on the WAN firewall zone.
- **Firewall / ACLs:** The deployment of inter-VLAN traffic filtering policies entails constraining Guest network access to internal resources while permitting solely the necessary services to communicate with the Server segment, designated as VLAN 20. Within the OpenWrt environment, these security measures are operationalized through the configuration of zones, forwarding rules, and specific directives located within the /etc/config/firewall file.

Updated Physical Topology The following diagram illustrates the active Packet Tracer topology, labeling both sites, their active LAN subnets, the WAN link, and the physical interface connections currently implemented:



2. VLSM Addressing

To fulfill the project requirements and accommodate the estimated 135 simultaneous devices, the 10.10.0.0/16 network block has been subnetted using VLSM. The active segments for this initial simulation phase are the User networks (VLAN 10) and the intersite WAN link. The remaining subnets have been formally allocated and reserved in the addressing plan to preserve the planned segmentation for completion after the Week 10 switching content (Router-on-a-Stick integration).

Subnet Allocation Table (Derived from 10.10.0.0/16)

Purpose / Location	Host Req.	Network Address	Prefix / Subnet Mask	Usable Host Range	Broadcast Address	Gateway	Allocation	Status
Sede A - VLAN 10 (Users)	126	10.10.10.0	/25 (255.255.255.128)	10.10.10.1 - 10.10.10.126	10.10.10.127	10.10.10.1	DHCP	Active
Sede A - VLAN 20 (Servers)	30	10.10.20.0	/27 (255.255.255.224)	10.10.20.1 - 10.10.20.30	10.10.20.31	10.10.20.1	Static	Planned
Sede A - VLAN 30 (Guests)	62	10.10.30.0	/26 (255.255.255.192)	10.10.30.1 - 10.10.30.62	10.10.30.63	10.10.30.1	DHCP	Planned
Sede A - VLAN 99 (Mgmt)	14	10.10.99.0	/28 (255.255.255.240)	10.10.99.1 - 10.10.99.14	10.10.99.15	10.10.99.1	Static	Planned
Sede B - VLAN 10 (Users)	62	10.10.110.0	/26 (255.255.255.192)	10.10.110.1 - 10.10.110.62	10.10.110.63	10.10.110.1	DHCP	Active
Sede B - VLAN 30 (Guests)	30	10.10.130.0	/27 (255.255.255.224)	10.10.130.1 - 10.10.130.30	10.10.130.31	10.10.130.1	DHCP	Planned

Sede B - VLAN 99 (Mgmt)	14	10.10.199.0	/28 (255.255.255.240)	10.10.199.1 - 10.10.199.14	10.10.199.15	10.10.199.1	Static	Planned
WAN Link (Inter-site)	2	10.10.254.0	/30 (255.255.255.252)	10.10.254.1 - 10.10.254.2	10.10.254.3	N/A	Static	Active

3. Basic Device Configuration

Device Configuration & Connectivity Basic configurations have been applied to the primary edge routers (ISR 4331) and distribution switches (Catalyst 2960) at both sites to establish initial Layer 3 addressing and Layer 2 connectivity. Hostnames have been set, and the physical interfaces for the currently active User networks (VLAN 10 simulations) and the WAN link have been assigned their respective IP addresses from the VLSM table. Configurations have been successfully saved to the startup-config to ensure persistence.

Figure 1

Ip interface of R1-SedeA.

```
R1-SedeA#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.10.10.1      YES manual up                    up
GigabitEthernet0/0/1   10.10.254.1     YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
R1-SedeA#
```

Self-made (2026).

Figure 2

Ip interface of R2-SedeB.

```
R2-SedeB#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.10.110.1     YES manual up                    up
GigabitEthernet0/0/1   10.10.254.2     YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
R2-SedeB#
```

Self-made (2026).

Figure 3

Ip interface of SW1-SedeA.

```
SW1-SedeA#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  10.10.10.2      YES manual up                    up
SW1-SedeA#
```

Self-made (2026).

Figure 4

Ip interface of SW2-SedeB.

```
SW2-SedeB#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  10.10.110.2     YES manual up                    up
SW2-SedeB#
```

Self-made (2026).

Figure 5

MAC address table of SW1-SedeA.

```
SW1-SedeA#ping 10.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/7 ms

SW1-SedeA#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.c727.e901    DYNAMIC     Fa0/1
SW1-SedeA#
```

Self-made (2026).

Figure 6

MAC address table of SW2-SedeB.

```
SW2-SedeB#ping 10.10.110.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.110.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

SW2-SedeB#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0003.e4ce.8401    DYNAMIC     Fa0/1
SW2-SedeB#
```

Self-made (2026).

Configuration applied to Sede A Router (R1-SedeA):

```
enable
 configure terminal
 hostname R1-SedeA
 interface GigabitEthernet0/0/0
  description LAN_SedeA_VLAN10
  ip address 10.10.10.1 255.255.255.128
  no shutdown
  exit
 interface GigabitEthernet0/0/1
  description WAN_Link_To_SedeB
  ip address 10.10.254.1 255.255.255.252
  no shutdown
  exit
 end
 write memory
```

Configuration applied to Sede B Router (R2-SedeB):

```
enable
 configure terminal
 hostname R2-SedeB
 interface GigabitEthernet0/0/0
  description LAN_SedeB_VLAN10
  ip address 10.10.110.1 255.255.255.192
  no shutdown
  exit
 interface GigabitEthernet0/0/1
```

```
 description WAN_Link_To_SedeA
 ip address 10.10.254.2 255.255.255.252
 no shutdown
 exit
end
write memory
```

Configuration applied to SW1-SedeA:

```
 enable
 configure terminal
 hostname SW1-SedeA
 interface vlan 1
  ip address 10.10.10.2 255.255.255.128
  no shutdown
  exit
 ip default-gateway 10.10.10.1
 end
 write memory
```

Configuration applied to SW2-SedeB:

```
 enable
 configure terminal
 hostname SW2-SedeB
 interface vlan 1
  ip address 10.10.110.2 255.255.255.192
  no shutdown
  exit
 ip default-gateway 10.10.110.1
 end
 write memory
```

MAC Address Table Analysis

To confirm the status of active interfaces and establish fundamental Layer 2 connectivity, each switch successfully executed a ping test to its local gateway via its management interface (VLAN 1), yielding a perfect success rate of 5 out of 5 replies as illustrated in Figures 5 and 6.Subsequently, the `show mac address-table` command revealed dynamic VLAN 1 entries corresponding to the MAC address of the router's GigabitEthernet0/0/0 interface: specifically, 0001.c727.e901 learned on port Fa0/1 at SW1-SedeA, and 0003.e4ce.8401 learned on the equivalent port at SW2-SedeB.

It is important to note that switches populate these address table entries dynamically by extracting source MAC addresses from incoming frames; consequently, entries for client PCs connected to port Fa0/2 are acquired through the identical mechanism whenever such devices transmit traffic. This verification confirms that the physical connections are operational, the interfaces have transitioned to an up state, and the switches execute dynamic learning procedures prior to forwarding frames out of the designated ports.

4. DHCP Configuration

4.1. Pool designs

A DHCP pool was configured on each router to automatically assign IP addresses to devices on the user network at each location. Since the routers function as gateways for their respective LANs, they can assign addresses directly, without the need to configure a DHCP relay service.

At this time, the Guest, Management, and Server networks do not have DHCP pools, as their addresses are reserved until VLAN separation is implemented. Thus, automatic assignment applies only to the networks used in testing.

In addition, the first ten IP addresses (.1 through .10) were reserved in both pools. This ensures that addresses for the gateway, switch management, and other equipment requiring a static IP remain available. This prevents DHCP from assigning those addresses to other devices and avoids network conflicts.

Figure 7

DHCP of R1-SedeA.

```
R1-SedeA#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.10.10.11      0004.9A22.7030           --                     Automatic
R1-SedeA#
```

Self-made (2026).

Figure 8

DHCP of R2-SedeB.

```
R2-SedeB#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.10.110.11     0002.168B.5030           --                     Automatic
R2-SedeB#
```

Self-made (2026).

Figure 9

DHCP Section of R1-SedeA.

```
R1-SedeA#show running-config | section dhcp
ip dhcp excluded-address 10.10.10.1 10.10.10.10
ip dhcp pool LAN_SEDEA
 network 10.10.10.0 255.255.255.128
 default-router 10.10.10.1
 dns-server 8.8.8.8
R1-SedeA#
```

Self-made (2026).

Figure 10

DHCP Section of R2-SedeB.

```
R2-SedeB#show running-config | section dhcp
ip dhcp excluded-address 10.10.110.1 10.10.110.10
ip dhcp pool LAN_SEDEB
 network 10.10.110.0 255.255.255.192
 default-router 10.10.110.1
 dns-server 8.8.8.8
R2-SedeB#
```

Self-made (2026).

Pool summary

LAN_SEDEA (R1-SedeA): network 10.10.10.0/25, excluded addresses 10.10.10.1–10.10.10.10, default gateway 10.10.10.1, DNS 8.8.8.8; PC0 received 10.10.10.11. LAN_SEDEB (R2-SedeB): network 10.10.110.0/26, excluded addresses 10.10.110.1–10.10.110.10, default gateway 10.10.110.1, DNS 8.8.8.8; PC1 received 10.10.110.11. Only these two user networks are claimed as tested; Guest, Server and Management pools are pending VLAN segmentation.

5. Initial intersite routing

Intersite connectivity between Sede A and Sede B has been configured via static routes across the 10.10.254.0/30 Wide Area Network (WAN) interface. The deployment of static routing for this initial phase was justified by the network's limited scope, comprising merely two sites and a solitary path, thereby ensuring a configuration that is straightforward, predictable, and readily verifiable. In this setup, each router designates the opposing endpoint of the /30 link as its next hop. These static routes are scheduled to be superseded by OSPFv2 upon the completion of the VLAN segmentation and Router-on-a-Stick implementation phases.

A static route configured on R1-SedeA provides connectivity to the Sede B LAN:

ip route 10.10.110.0 255.255.255.192 10.10.254.2

Static route on R2-SedeB (reaching sede A's LAN):

ip route 10.10.10.0 255.255.255.128 10.10.254.1

Figure 11

Routing table of R1-SedeA

```
R1-SedeA#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
C       10.10.10.0/25 is directly connected, GigabitEthernet0/0/0
L       10.10.10.1/32 is directly connected, GigabitEthernet0/0/0
S       10.10.110.0/26 [1/0] via 10.10.254.2
C       10.10.254.0/30 is directly connected, GigabitEthernet0/0/1
L       10.10.254.1/32 is directly connected, GigabitEthernet0/0/1

R1-SedeA#
```

Self-made (2026).

Figure 12

Routing table of R2-SedeB

```
R2-SedeB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

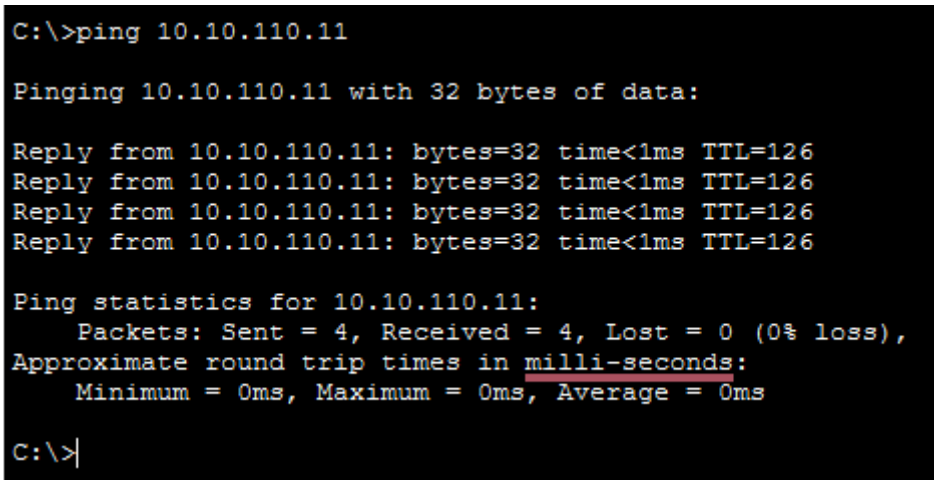
     10.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
S       10.10.10.0/25 [1/0] via 10.10.254.1
C       10.10.110.0/26 is directly connected, GigabitEthernet0/0/0
L       10.10.110.1/32 is directly connected, GigabitEthernet0/0/0
C       10.10.254.0/30 is directly connected, GigabitEthernet0/0/1
L       10.10.254.2/32 is directly connected, GigabitEthernet0/0/1

R2-SedeB#
```

Self-made (2026).

Figure 13

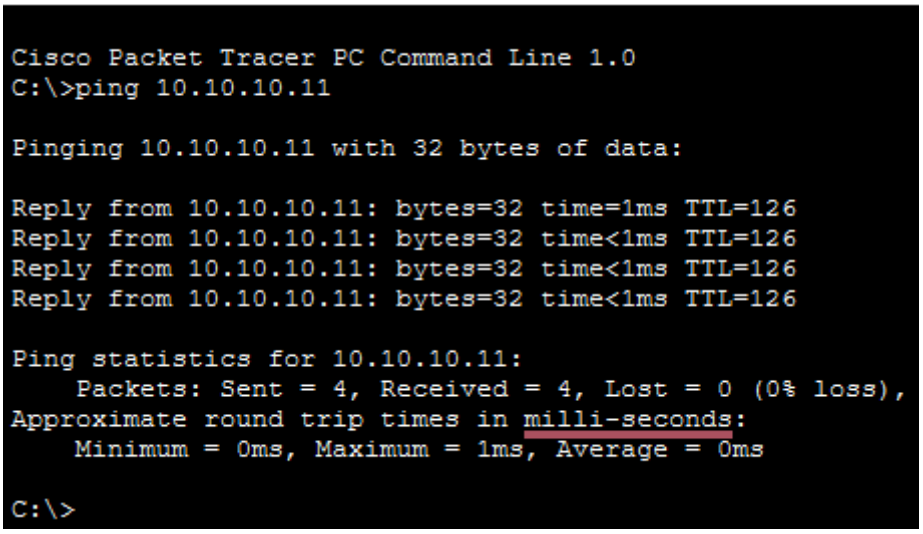
Ping from PC0 to PC1



Self-made (2026).

Figure 14

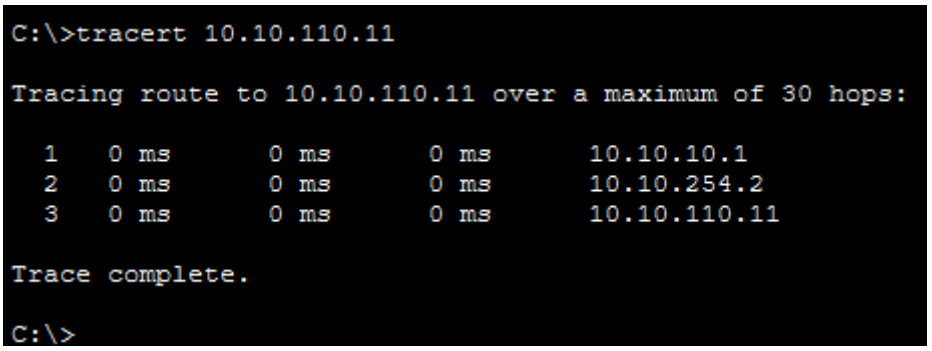
Ping from PC1 to PC0



Self-made (2026).

Figure 15

Tracert of PC0 to PC1



Self-made (2026).

6. Administration and Troubleshooting

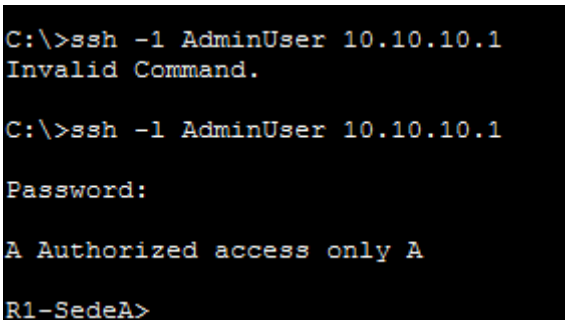
Secure administration protocols have been rigorously implemented across both routers and switches, establishing a multi-layered defense.Access to Privileged EXEC mode is safeguarded by an encrypted enable secret, while the service password-encryption command ensures that all stored passwords are obfuscated.

Console line access is restricted via a dedicated password, complemented by a login banner that explicitly warns of the consequences for unauthorized intrusion. Remote management is exclusively facilitated through SSH version 2 utilizing local user authentication; furthermore, the VTY lines are configured to accept only SSH traffic (transport input ssh), thereby completely disabling the insecure Telnet protocol.

The switches currently host management addresses on VLAN 1, specifically 10.10.10.2/25 and 10.10.110.2/26, which were deliberately assigned from address ranges excluded from DHCP pools to prevent conflicts. Upon the completion of network segmentation, these management interfaces will be migrated to VLAN 99. To ensure confidentiality, this report and the accompanying repository contain no credentials, and all configuration backups have undergone a thorough sanitization process prior to upload.

Figure 16

SSH active (test from PC0).



Self-made (2026).

Troubleshooting method

Problems were diagnosed following the OSI model bottom-up, using the codes below.

Layer	Command	Purpose
Physical (1)	*show ip interface brief*	Interface status
Data Link (2)	*show mac address-table*	MAC on the switches
Network (3)	*show ip route, ping, tracert*	Routing and reachability
Services	*show ip dhcp binding, show ip ssh*	DHCP leases and SSH status

Figure 17

SSH enabled.

```
R1-SedeA#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1-SedeA#
```

Self-made (2026).

7. Preliminary Testing Matrix

The following matrix stores the tests executed on the Packet Tracer prototype. Features that depend on later phases cannot be tested as of now and are marked as Pending.

ID	Source	Destination	Procedure	Expected result	Observed result	Evidence	Status
T1	R1-SedeA	Local interfaces	show ip interface brief	Gi0/0/0 and Gi0/0/1 up/up	10.10.10.1 and 10.10.254.1, up/up	Fig. 1	Pass
T2	R2-SedeB	Local interfaces	show ip interface brief	Gi0/0/0 and Gi0/0/1 up/up	10.10.110.1 and 10.10.254.2, up/up	Fig. 2	Pass
T3	SW1-SedeA	Local interfaces	show ip interface brief	Fa0/1, Fa0/2, Vlan1 up/up	Vlan1 10.10.10.2 up/up; Fa0/1, Fa0/2 up/up	Fig. 3	Pass
T4	SW2-SedeB	Local interfaces	show ip interface brief	Fa0/1, Fa0/2, Vlan1 up/up	Vlan1 10.10.110.2 up/up; Fa0/1, Fa0/2 up/up	Fig. 4	Pass
T5	SW1-SedeA	10.10.10.1	ping	0% loss	5/5 replies	Fig. 5	Pass
T6	SW2-SedeB	10.10.110.1	ping	0% loss	5/5 replies	Fig. 6	Pass
T7	SW1 / SW2	Local	show mac address-table	Dynamic entry for router Gi0/0/0	Learned on Fa0/1 (0001.c727.e901 / 0003.e4ce.8401)	Figs. 5, 6	Pass
T8	PC0	R1-SedeA (DHCP)	DHCP request	Address in 10.10.10.11–126	10.10.10.11	Figs. 7, 9	Pass
T9	PC1	R2-SedeB (DHCP)	DHCP request	Address in 10.10.110.11–62	10.10.110.11	Figs. 8, 10	Pass
T10	R1 / R2	Routing tables	show ip route	Static route to the remote LAN	S 10.10.110.0/26 via 10.10.254.2; S 10.10.10.0/25 via 10.10.254.1	Figs. 11, 12	Pass
T11	PC0	PC1 (10.10.110.11)	ping	0% loss	4/4 replies, TTL=126	Fig. 13	Pass

T12	PC1	PC0 (10.10.10.11)	ping	0% loss	4/4 replies, TTL=126	Fig. 14	Pass
T13	PC0	PC1	tracert	Path through both routers	10.10.10.1, 10.10.254.2, 10.10.110.11	Fig. 15	Pass
T14	PC0	R1-SedeA	SSH login	Login prompt, banner, CLI access	Login successful, banner shown	Fig. 16	Pass
T15	PC0 / PC1	R2-SedeB, SW1, SW2	SSH login	Login successful	[fill]	Fig. 17	[Pass / Pending]
T16	PCs	VLANs 10/20/30/99, trunks, Router-on-a-Stick	To be defined	Inter-VLAN connectivity	Not implemented	N/A	Pending
T17	Switches	EtherChannel (LACP), RSTP	To be defined	Aggregated link, loop-free topology	Not implemented	N/A	Pending
T18	R1 / R2	OSPFv2 adjacency	To be defined	Neighbor FULL, routes learned via OSPF	Not implemented	N/A	Pending
T19	Routers	NAT and firewall rules	To be defined	Translation and filtering as designed	Not implemented	N/A	Pending
T20	VLAN 20	Go (Email) and FastAPI (Auth) servers	To be defined	Services reachable from VLAN 10	Not implemented	N/A	Pending

8. Completion and OpenWrt Migration Plan

Subsequent tasks have been delineated and aligned with the Week 10 equipment delivery schedule.Cisco Command-Line Interface directives cannot be directly ported to OpenWrt; consequently, every function must be reconstructed using proprietary UCI configuration files and distinct syntax rather than being merely translated or copied.

Task	Owner	Target	Depends on
VLANs 10/20/30/99 and access ports	Paula	Week 10	Basic LAN
802.1Q trunks and Router-on-a-Stick	Paula	Week 10	VLANs
EtherChannel (LACP) and RSTP	Odalis	Week 10–11	VLANs
OSPFv2 (replaces static routes)	Odalis	Week 11	Router-on-a-Stick
NAT (PAT) toward the upstream link	Keyla	Week 11	Routing
Firewall / ACLs between VLANs	Keyla	Week 11	VLANs, routing
Go and FastAPI servers in VLAN 20	Keyla	Week 11	VLAN 20
OpenWrt installation and UCI configuration	All	Week 11	Equipment delivery
Re-run the testing matrix on hardware	All	Week 11	OpenWrt configuration

Cisco concept to OpenWrt equivalent

Function	Cisco concept	OpenWrt equivalent
Interface addressing	ip address on the interface	config interface with proto 'static', ipaddr, netmask in /etc/config/network
DHCP	Pool with excluded addresses	config dhcp in /etc/config/dhcp; the excluded range is expressed with start and limit
Static routes	ip route	config route in /etc/config/network
VLANs / Router-on-a-Stick	Trunk and dot1q subinterfaces	802.1Q devices (config device, option vid) on the physical port
Dynamic routing	OSPF process	OSPF daemon package (e.g., BIRD or FRR)
NAT	ip nat inside/outside	masq option on the WAN firewall zone
Filtering	ACLs	Zones, forwardings and rules in /etc/config/firewall
Remote access	SSH v2 with local users	Dropbear (/etc/config/dropbear)
Saving	write memory	uci commit and service reload
EtherChannel / RSTP	Switch features	Kept on the managed switches; OpenWrt alternatives to be evaluated on the hardware

Example not yet tested in Sede A:

```
# /etc/config/network

config interface 'lan'

    option device 'br-lan'

    option proto 'static'

    option ipaddr '10.10.10.1'

    option netmask '255.255.255.128'


config interface 'wan_link'

    option device 'eth1'

    option proto 'static'

    option ipaddr '10.10.254.1'

    option netmask '255.255.255.252'


config route

    option interface 'wan_link'

    option target '10.10.110.0/26'

    option gateway '10.10.254.2'


# /etc/config/dhcp

config dhcp 'lan'

    option interface 'lan'

    option start '11'

    option limit '116'

    option leasetime '12h'
```

For sede B, the equivalents are:

10.10.110.1/255.255.255.192, a route to 10.10.10.0/25 via 10.10.254.1, and a pool with start '11' and limit '52'.

Migration steps

1. Ensure that the sanitized Packet Tracer configurations are duly backed up.
2. Deploy OpenWrt onto the hardware acquired during Week 10.
3. Configure interfaces, DHCP and static routes with UCI and commit.
4. Establish the Wide Area Network connection linking the two facilities and subsequently validate the link's integrity using a ping test.
5. Execute the testing matrix across the hardware platform once more and document any observed discrepancies.
6. Proceed to implement VLANs, OSPF routing, NAT, and firewall rules in the sequence specified by the work table.

Updated schedule

- Week 9: Packet Tracer prototype (addressing, DHCP, static routing, SSH).
- Week 10: equipment delivery, VLANs, trunks and Router-on-a-Stick.
- Week 11: complete simulation and initial hardware deployment.
- Week 15 serves as the culminating review.
- Week 16 concludes with the final presentation.

Risks

- Equipped with only a solitary Ethernet port, the Raspberry Pi necessitates the integration of a USB-Ethernet adapter to establish a secondary network interface.
- The nomenclature of network interfaces, such as eth0 and eth1, requires confirmation through inspection of the actual hardware.
- EtherChannel and RSTP depend on managed switches.
- The implementation of OSPF necessitates the installation of a supplementary routing package.

Team contributions

Team Member	Contributions
Paula Cevallos	Providing technical coordination, compiling and authoring the comprehensive report, and drafting Sections 6 through 8, which cover administration and troubleshooting, the testing matrix, and the migration plan, followed by conducting the final review of the entire document.
Odalis Anangonó	Sections 3 through 5 encompass the foundational device setup, the implementation of DHCP parameters, and the establishment of initial inter-site routing protocols, accompanied by corresponding illustrative screenshots.
Keyla Imba	This deliverable encompasses the revised network architecture detailed in Section 1, the Variable Length Subnet Masking (VLSM) addressing scheme outlined in Section 2, and the corresponding device configurations executed within Cisco Packet Tracer.

9. AI Declaration

During this phase, the team leveraged Claude (Anthropic) as a scholarly and drafting instrument.Its functions encompassed elucidating Cisco IOS command syntax, directing configuration procedures for DHCP, static routing, and SSH, drafting textual content for Sections 6 through 8 alongside the OpenWrt migration strategy, and auditing the document's internal consistency.

Notwithstanding these auxiliary tools, every configuration was manually inputted and rigorously verified by the team within Cisco Packet Tracer; all figures presented are original screenshots generated by the team itself.Furthermore, the team conducted a comprehensive review and editorial revision of the material, assuming full responsibility for the final content.

External Resources

Cisco Packet Tracer, a network simulation tool provided by the Cisco Networking Academy, was employed for this exercise.

Consultation of the Cisco IOS command documentation available on cisco.com was conducted to ascertain the precise configuration syntax.

Reference was made to the OpenWrt documentation (openwrt.org/docs) to inform the UCI, network, DHCP, and firewall configurations integral to the migration strategy.